Example 2.8 Kannan divides a circular disc of radius 14 cm into four equal parts. What is the perimeter of a quadrant shaped disc? (use $\pi = \frac{22}{7}$)

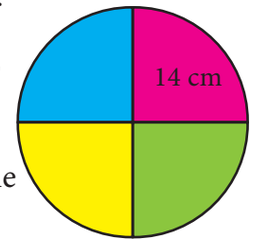


Fig. 2.10

Solution

To find the perimeter of the quadrant disc, we need to find the circumference of quadrant shape.

Given that radius (r) = 14 *cm*.

We know that the circumference of circle $= 2\pi r$ *units*.

So, the circumference of the quadrant arc $= \frac{1}{4} \times 2\pi r$

$= \frac{\pi r}{2}$

$= \frac{22}{7} \times \frac{14}{2}$

$= 22$ *cm*

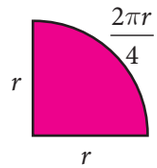


Fig. 2.11

Given, the radius of the circle $= 14$ *cm*

Thus, perimeter of required quadrant shaped disc $= 14 + 14 + 22$

$= 50$ *cm*.

Think

(i) Is the circumference of the semicircular arc and semicircular shaped disc same? Discuss.

(ii) The traffic lights are circular. Why?

(iii) When you throw a stone on still water in pond, ripples are circular. Why?

Exercise 2.1

1. Find the missing values in the following table for the circles with radius (r), diameter (d) and Circumference (C).

S.No.	radius (r)	diameter (d)	Circumference (C)
(i)	15 cm		
(ii)			1760 cm
(iii)		24 m	

2. Diameters of different circles are given below. Find their circumference (Take $\pi = \frac{22}{7}$).

   (i) $d = 70$ *cm* (ii) $d = 56$ *m* (iii) $d = 28$ *mm*

3. Find the circumference of the circles whose radii are given below.

   (i) 49 *cm* (ii) 91 *mm*

4. The diameter of a circular well is 4.2 *m*. What is its circumference?

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